

DS227 Business Analytics For Data Science

Final Project Report: Customer Segmentation Analysis for a UK Online Retailer

Aleks Mantashyan

1. Introduction And Purpose Of The Project

This project analyzes one year of transaction data from a London-based online retailer company specializing in household items and gifts. The goal of the project is to understand and gain insights about customers' behavior, identify and highlight high-value segments, and having all this information, understand if targeted marketing can be used to increase sales and, in general, improve the company's performance.

2. Business strategy and Data Description

As we know from the weighting of RFM attributes, online retail companies pay more attention to and depend heavily on repeat customers rather than one-time buyers, even if the monetary value is large. Knowing this, this company should have a strong interest in understanding how customer value evolves and which customers are most worth retaining.

The main research question is:

- How can the company identify and retain high-value customers to increase lifetime value and the revenue the company produces?

Supporting questions include:

- Which customer segments produce the most revenue for the company?
- How do purchase frequency, spending, and recency vary across customers?
- Can we predict which customers are likely to be high-value?
- Do personalized offers improve conversions compared to generic ones?

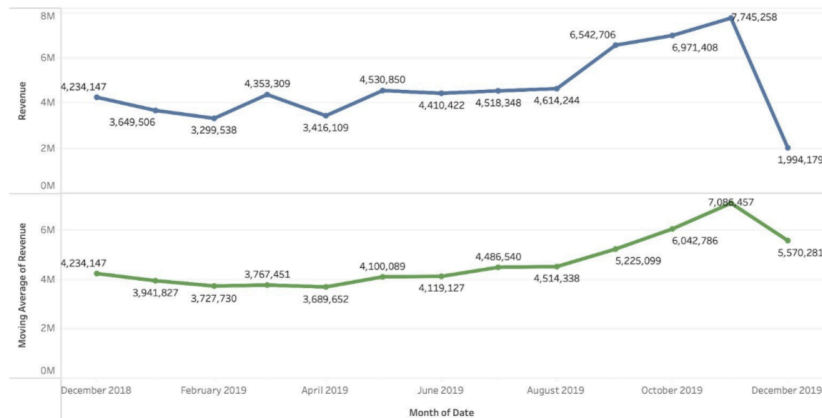
The "sales_transaction.csv" dataset from kaggle that we are using contains more than 500,000 rows. Each of the rows represents a purchased product. Variables include

- Transaction number
- Date
- Product
- Price
- Quantity
- Customer ID
- Country

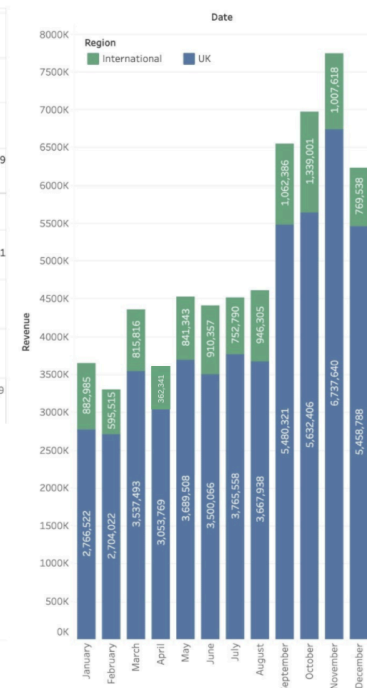
After removing cancelled orders, negative quantities and prices, and rows that do not have customer IDs, we get a cleaned dataset that provides a reliable foundation for customer analysis.

Retail Revenue Insights: Trends, Regional Performance and Top Products

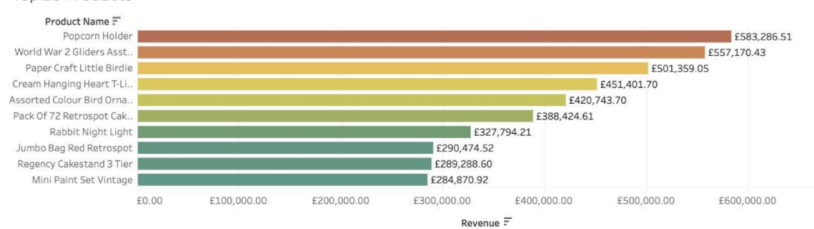
Revenue Over Time



UK VS International Revenue



Top 10 Products



The Tableau dashboard shows important revenue trends and product performance during 2018 and 2019. The upper left part shows the revenue produced by the company over the months. We can see slight fluctuations during the first half of the year, but starting from August we can see that it rises, reaching a peak of 7.75 million pounds in December. This is very natural when we look from the business perspective, because people tend to buy more gifts and household items before the holidays.

The lower left part shows top 10 products that generate the most revenue for the company, those are mostly giftware and decorative items. The best selling product is a popcorn holder, which generated over 580.000 pounds. Knowing this, the company can concentrate on selling items similar to those.

The left part shows regional breakdown which makes it obvious that the UK accounts for the majority of the total revenue, which is natural since the company is based in London. Despite this, international buyers also have meaningful shares especially during the later half of the year.

The main business strategy used for this project is customer value optimization. The company's goal is to strengthen retention and increase the lifetime value of the customers it already has. Since acquiring a new customer is connected with higher costs than retaining existing customers, this approach is more cost effective.

To support these goals, the analysis uses KPIs such as:

- Total revenue = Price * Quantity
- Average order value
- Purchase frequency
- Repeat rate and churn rate
- Customer Lifetime Value
- RFM scores

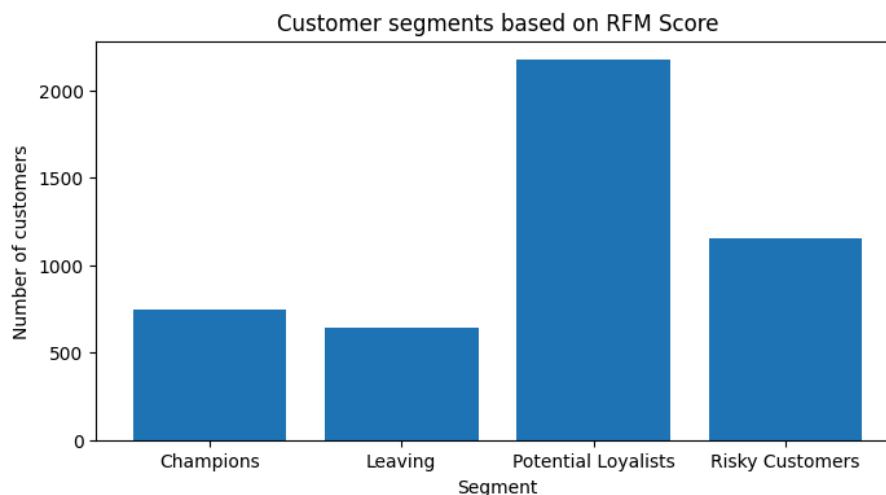
These metrics help describe customer behavior and serve as inputs for clustering, classification, CLV estimation, and A/B testing.

3. Descriptive Analysis and RFM Model

To begin analyzing customer behavior, I calculated RFM values for each customer:

- Recency: days since last purchase
- Frequency: number of unique purchases
- Monetary: total money spent

I transformed those into ranks and normalized them from 0 to 100 in a way that for example most recent customers received a high Recency rank. Ranking ensures that later, no single variable dominates the clustering methods that use distances.



The RFM segmentation shows clear differences in customer behavior. Potential Loyalists make up the largest group, with more than 2,000 customers who purchase fairly often and spend consistently. Risky Customers form the next largest segment, indicating a sizable group at risk of churning. Champions represent around 750 of the most valuable and engaged customers, while the Leaving segment containing about 650 customers includes buyers who have not purchased for a long time. This distribution highlights strong growth potential but also the importance of re-engaging at-risk customers.

4. Customer Segmentation with K-Means

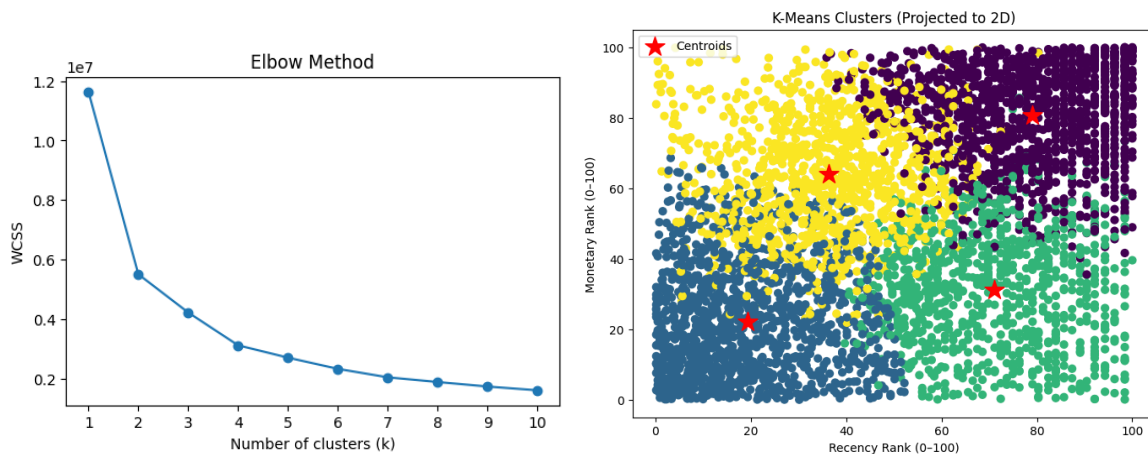
K-Means clustering is used to separate groups of customers with similar purchasing patterns with no predefined labels. This reveals naturally occurring segments that otherwise could be hidden when only using summary statistics

The normalized RFM scores made an ideal input because they focus purely on customer behavior, place all features on the same scale and support distance-based algorithms like K-Means

When doing K-Means it is important to get an optimal number k which shows the number of clusters, so I evaluated models with $k = 2$ to 10 clusters using:

- Elbow method
- Silhouette score

The elbow plot shows the line “breaking” at $k = 4$, and the silhouette score for $k = 4$ (0.378) was significantly stronger than $k = 3$ (0.342) while maintaining meaningful interpretability. Despite the fact that $k = 2$ had the highest silhouette score, using $k = 2$ would get us very harsh segments. Therefore, $k = 4$ was selected as the optimal balance.



K-Means produced four reasonably balanced groups:

- Cluster 0 - Top Tier Customers/Champions: 1310 customers
- Cluster 1 - At-Risk Customers: 1353 customers
- Cluster 2 - New Customers/Occasional Buyers: 942 customers
- Cluster 3 - Potential Loyalists: 1113 customers

Their behavioral patterns were:

- Cluster 0: customers who buy frequently, spend the most money, and purchased recently. So this cluster includes the company’s most profitable group.
- Cluster 3: considerable engagement customers who have good frequency and spending; those kinds of customers have strong growth potential.
- Cluster 2: customers who buy from time to time with low purchasing amounts.
- Cluster 1: very inactive customers who have not purchased in a long time.

It was expected that the distribution would be uneven. Customers in cluster 0 generated about 71.5% of total revenue, and cluster 3 another 20.4%. Together, these two segments generated more than 90% of total sales.

These two groups generate a very big part of the total revenue, so I decided to test a scenario where the company increases revenue from Clusters 0 and 3 by 5%. As expected, this results in approximately increasing the revenue by 2.89 million, which would make the revenue to reach about

60.75 million. This suggests that targeted marketing towards loyal and potential-loyal customers could significantly increase returns.

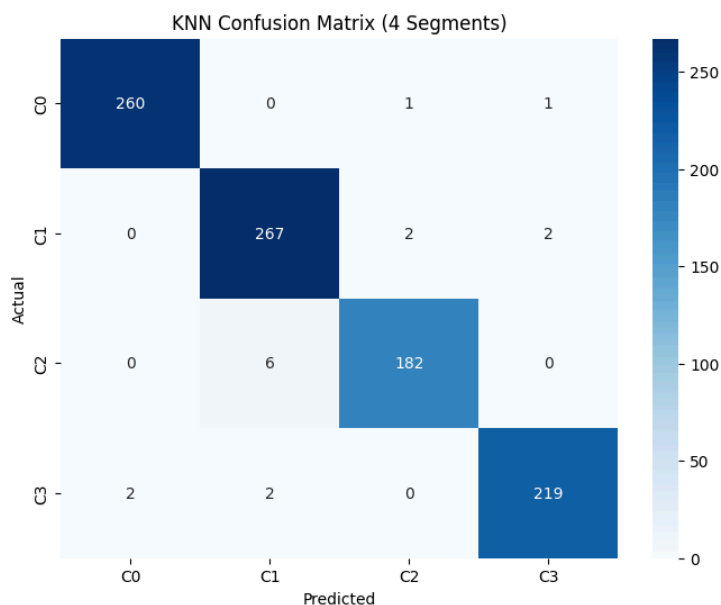
5. KNN Classification

I trained a K-Nearest Neighbors (KNN) classifier using the RFM ranks that I created and normalized earlier. After splitting the dataset into training and test sets (80/20) and applying StandardScaler, I trained a KNN model with $k = 5$.

After checking if the model works well, I got the following results:

- Accuracy: 0.9831
- Precision: 0.9834
- Recall: 0.9819
- F1 Score: 0.9826

Judging from these high results, we can say that RFM behavior is a strong indicator of which segment a customer belongs to, and it's useful for scoring customers.



The Confusion Matrix shows that KNN classifier performs very well across all four customer segments. There is a strong agreement between actual and predicted labels, as we can see from the diagonal. There are only a few misclassifications, e.g. only a few C1 customers were thought to be in C2 or C3, and similarly small errors appear between C0 and C3. So we can say confidently that the model captures the segmentation patterns in a reliable way, confirming that RFM features provide enough information to accurately classify customers.

6. Customer Lifetime Value (CLV)

I computed CLV using those components:

- Purchase frequency: 4.194
- Repeat rate: 0.6657
- Churn rate: 0.3343
- Profit margin: 10%

CLV was computed customer by customer and segmented into quartiles that ranged from Segment A, including the highest-value customers to Segment D, including the lowest-value ones. Segments B and C represented mid-value customers. While A and D show the strongest contrast, the full segmentation helps identify customers with moderate but still meaningful value potential.

The CLV distribution was very uneven, very similar to the spending distribution. A small group of customers accounts for most of the company's customers' total lifetime value.

7. Regression Analysis for CLV

Next, I tried to predict CLV using linear regression, but the model performed very poorly:

- MAE : 3470.161
- RMSE: 32364.207
- R^2 : 0.1729609

This was expected, because the nature of the analysis I was doing did not correspond to the kind of analysis that fits well with Linear Regression. CLV is heavily skewed and influenced by factors that are not linearly related to RFM.

8. Logistic Regression

Since linear regression was not suitable for CLV prediction, I tried to use regression in another way and try to use logistic regression for binary classification: distinguishing high- value customers included in clusters 0 and 3 from the low-value customers. Logistic regression performed reasonably well and confirmed that RFM features predict customer value reliably.

- Accuracy: 0.95127
- Precision (high value class): 0.95825
- Recall (high value class): 0.94639
- F1 Score (high value class): 0.95228

The results show an accuracy score of approximately 95%, proving the logistic regression does a good job at classifying most of the customers. Precision score is again about 95%, showing that when the model tells us that the customer is high-value, then in 95% of the cases the customer is actually high-value. The recall score is 0.94639, showing that the model successfully finds most of the true high-value customers. And the F1 score of 0.952 indicates that there is a strong balance between precision and recall This model can support targeted campaigns by identifying which customers may become high-value in the future.

9. A/B Testing

I did a simulation of an A/B test that compared a generic offer (control) to a personalized offer (treatment) to measure if personalization improves customer conversion. I used a baseline conversion rate of 12% and a minimum detectable effect of 5%. The analysis showed that approximately 47000 customers per group were required.

Outcomes:

Group	Conversions	Conversion Rate
Control	5,567	11.84%
Treatment	5,879	12.51%

The personalized offer increased conversions by 0.66 percentage points, representing a 5.6% relative lift over the generic offer.

Then it was needed to test whether this improvement was statistically significant, so I conducted a chi-square test and t-tests for proportions.

To test whether this improvement was statistically meaningful, both a chi-square test and a t-test for proportions were conducted. Each test resulted in a p-value of 0.002, which is less than the 0.05 threshold. This means we reject the null hypothesis and conclude that personalization leads to a significant increase in conversion.

A 95% confidence interval for the absolute lift ranged from 0.002 to 0.011, and the corresponding relative lift ranged from 2.1% to 9.1%. Even the lower bound of these intervals suggests a meaningful effect in practical business terms.

Overall, the A/B test provides strong evidence that personalized offers outperform generic ones and can materially improve conversion rates.

10. SWOT and Financial Analysis

To translate the analytical results into practical business implications, I conducted a SWOT analysis and connected the findings to basic financial reasoning.

Strengths: As we identified with K-Means, the company largely benefits from the customers in clusters 0 and 3. Those customers represent less than half of the customers base, but still generate over 90% of the total revenue. They buy frequently, spend a lot of money and are loyal to the company. This all gives the business a stable revenue stream. Another key strength is that customer behavior is highly predictable using simple RFM features. Both the KNN classifier and logistic regression performed very well, meaning that the company can reliably identify high-value customers and make clever marketing decisions.

Weaknesses: One major weakness the company has is the high number of one-time buyers, which contributes to a 33% churn rate. This forces the business to depend more heavily on acquiring new customers, which, as we discussed earlier, is not cost-efficient. Customers included in clusters 1 and 2

generate very little spending, so spending marketing resources on them may not be cost-effective. Additionally, the CLV regression model could not properly predict lifetime value because of its skewed distribution. This limits the company's ability to make predictions about revenue.

Opportunities: The company has many opportunities to increase growth rate. According to the analysis in the upper section, increasing spending from high-value customers by just 5% would generate an additional 2.89 million pounds in revenue. This is because, as we discussed, most part of the revenue comes from a small group of loyal customers, so even a small investment in those customers can have a very large and positive financial impact.

There is also potential to reduce churn, which directly increases CLV, and the A/B test demonstrated that personalized offers improve conversion in a significant way, suggesting that personalization could be expanded across other channels.

Threats: Although having a small, but loyal customer segment creates opportunities, there are also threats involved. Losing even a small number of those customers would significantly impact revenue. Other companies that offer similar products and services may pull those customers away. Rising acquisition costs further threaten profitability, especially for low-value customers.

To understand whether the company should use its resources to act on those insights, financial tools can be applied to make financial analysis.

The company could invest 500,000 points in a targeted campaign aimed at customers in Clusters 0 and 3. The company would get 2.89 million pounds which implies an extremely short payback period of about two months. Taking into account the 10% profit margin, the added profit would be around 289,000 pounds, giving an Accounting Rate of Return (ARR) of 58%, which is very impressive. Even when using a relatively high discount rate (10%), simple DCF reasoning shows that improving retention among high-value customers produces positive net present value.

So taking all those results together, we can confidently say that investing in high-value customer retention is financially safe and correct.

11. Key Findings, Recommendations, and Limitations

Key Findings

1. A small segment of customers generates most of the company's revenue. Even though customers in clusters 0 and 3 make up roughly half of the customer base, they generate over 90% of the total revenue. This indicates a highly uneven value distribution.
2. Customer behavior is predictable using simple RFM features.
Both the KNN and logistic regression models performed well. Considering recency, frequency and monetary patterns, high-value customers can be reliably identified.
3. Personalized marketing significantly improves conversions.
The A/B test demonstrated a statistically significant lift of 5.6%, with a 95% confidence interval that remained positive even at the lower bound. We can conclude that personalization is more effective than generic messaging.
4. Churn remains a major issue.
Approximately one-third of customers buy from the company only once. This not only limits long-term revenue potential but also increases reliance on new customer acquisition.

Recommendations

1. Start prioritizing high-value customers with targeted strategies that make the customer stay with the company
The company should put the focus on clusters 0 and 3 by offering personalized promotions, early-access deals or loyalty benefits. As shown, even small improvements in this group's engagement results in a large financial impact.
2. Start using predictive models to identify which customers can become high-value.
The logistic regression model can separate customers who can potentially become Champions and are beginning to show valuable behavior. Acting early can help the company retain them and avoid the situation in which they churn.
3. Stop spending heavily on low-value segments.
Clusters 1 and 2 contribute very little revenue. This should be reflected by the company's strategies. They should receive low-cost and automated communications instead of expensive marketing efforts.
4. Change acquisition and retention balance by reducing churn.
Improving churn by even a few percentage points would increase CLV across the entire customer base. Retention strategies should be expanded beyond campaigns. The company could start improving product recommendations or checkout reminders.

Limitations

- The dataset used for this analysis does not have demographic information, browsing behaviour, and marketing interaction data. Having this data could greatly improve segmentation and prediction.
- Simplified CLV calculation. CLV was estimated using assumptions about profit margin and churn that may not reflect real operational costs. This was again indirectly connected to the lack of data
- The A/B test was only a simulation, not a real-time experiment. Customer responses may differ in practice as it was not run in a live environment.
- RFM captures only past behaviour. RFM does not consider customer preferences or price sensitivity which could influence future purchases and therefore revenue.

What Additional Data or Analysis Would Strengthen Conclusions

- Customer demographics (age, location, income)
- Website behaviour data (sessions, clicks, abandoned carts)
- Marketing attribution data (email opens, ad impressions)
- Product category-level insights
- Running a real A/B test rather than a simulated one
- More advanced CLV models that include retention probability or time-series forecasting